Scope before Reuse: corrected theory sections (draft for review)

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Abstract

Corrected problem formulation, theory and related-work paragraph for FSE-2027-13; see docs/paper/THEORY.md for the derivation and docs/paper/latex/CHANGES.md for what each part replaces.

1 Problem Formulation

Skills, scopes and cells. A skill is $S = \langle \pi, \mathcal{D}, \mathcal{I}, \mathcal{P}, \mathcal{K} \rangle$: a procedure π (a parameterised edit program), dependency and environment fingerprints $\mathcal{D}$, input assumptions $\mathcal{I}$ (executable predicates on the target), observed postconditions $\mathcal{P}$ with the contexts they were observed in, and an interaction footprint $\mathcal{K}$, the code regions and behavioural properties that π reads or writes. A context x is a (repository, revision, environment, task) tuple. A scope family $\mathbb{S} = \{s_1, \ldots, s_L\}$ is a finite partition of contexts declared before any trial. We call a pair $c = (S, s)$ a *cell*; for skills S, S' with overlapping footprints, $(S \circ S', s)$ is a cell of its own.

Loss and selective transfer risk. For a task x, $\ell(x, \pi) \in [0, 1]$ is its functional loss with the skill applied and $\ell(x, \varnothing)$ the loss without it. We use one minus the held-out pass rate plus 0.5 per regression, *clipped at* 1, so that the range assumption below holds. The paired difference is $Z = \ell(x, \pi) - \ell(x, \varnothing) \in [-1, 1]$.

Definition 1.1 (Selective transfer risk). For a cell $c = (S, s)$ and a distribution P_s over the contexts of scope s, $\mathrm{STR}_P(c) = \mathbb{E}_{x \sim P_s}\big[\ell(x, \pi) - \ell(x, \varnothing)\big]$. A cell is *admissible* if $\mathrm{STR}_P(c) \le -\theta$ for a benefit margin $\theta > 0$, and *harmful* if $\mathrm{STR}(c) > 0$.

The definition is an excess over the no-skill baseline, so a skill that neither helps nor harms is not admissible, and it is relative to a declared scope; Proposition 2.4 shows why it cannot be otherwise.

Assumptions.

Assumption 1 (Paired trials). *For every cell c there is an infinite sequence of tasks $x_1^c, x_2^c, \ldots$ drawn i.i.d. from P_s (in practice, drawn* with replacement *from the target's task set); each task is run with and without the skill on the same snapshot and seed. Different cells may depend on each other.*

Assumption 2 (Bounded shift). *Deployment contexts of scope s follow Q_s with $\mathrm{TV}(P_s, Q_s) \le \xi_s$, where $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$.*

Admission problem. Given a library, a validation budget and a pre-allocated error budget α, decide which cells may execute, such that $\Pr[\exists \text{ an admitted cell } c : \mathrm{STR}_Q(c) > 0] \le \alpha$ over the whole library, over adaptively chosen cells and over re-admissions after retirement. Theorem 2.2 gives this guarantee whenever $2\xi_s < \theta$ for every scope; for coarser scopes it gives the weaker, explicit statement $\mathrm{STR}_Q(c) \le -\theta + 2\xi_s$.

2 Theory

We claim no new statistics: the guarantee is assembled from betting e-processes [9], Ville's inequality [6] and a union bound. What is new is the estimand, the per-cell ledger that covers compositions and re-admission, the proof that the shift term is unavoidable, and the exact characterisation of composition risk.

2.1 Per-cell evidence

Lemma 2.1 (Betting e-process). *Fix a cell c and let $Y_i = -Z_i$. Under $H_0^c : \mathbb{E}[Y] \le \theta$ (the cell is not admissible), the process $K_0 = 1$, $K_n = \prod_{i \le n}\big(1 + \lambda_i(Y_i - \theta)\big)$ with predictable $\lambda_i \in [0, 1/(1+\theta))$ satisfies $\Pr[\exists n : K_n \ge 1/\beta] \le \beta$ for every $\beta \in (0, 1)$.*

Proof. Since $Y_i \ge -1$, every factor is at least $1 - \lambda_i(1+\theta) > 0$. By Assumption 1 and predictability, $\mathbb{E}[K_i \mid \mathcal{F}_{i-1}] = K_{i-1}\big(1 + \lambda_i(\mathbb{E}[Y_i] - \theta)\big) \le K_{i-1}$, so (K_n) is a nonnegative supermartingale with $K_0 = 1$, and Ville's inequality gives the claim. □

The bound holds at any data-dependent stopping time, including the end of the validation budget, which is why a fixed-sample interval would not do.

2.2 Library-wide admission guarantee

Theorem 2.2 (Admission over a declared scope family). *Let C be the set of cells, compositions included, and let a ledger $\alpha_{c,k} \ge 0$ be fixed in advance for every cell c and epoch $k \ge 0$ (epoch $k \ge 1$ is the k-th re-admission after retirement), with $\sum_{c \in C} \sum_{k \ge 0} \alpha_{c,k} \le \alpha$; for instance $\alpha_{c,k} = \alpha_c \cdot 6/(\pi^2 (k+1)^2)$ with $\sum_c \alpha_c \le \alpha$. Suppose each epoch uses fresh tasks and admits c only when its e-process (Lemma 2.1) reaches $1/\alpha_{c,k}$, while the choice of cells, their order and the stopping times may depend on all data. Under Assumptions 1 and 2,*

$$\Pr\big[\exists \text{ admitted } (c, k) : \mathrm{STR}_Q(c) > -\theta + 2\xi_{s(c)}\big] \le \alpha.$$

In particular, if $2\xi_s < \theta$ for every scope, the probability of admitting any harmful cell is at most α.

Proof. For every (c, k) let $A_{c,k} = \{\sup_n K_n^{c,k} \ge 1/\alpha_{c,k}\}$, defined on the whole infinite task sequence of that epoch. An admission of (c, k) implies $A_{c,k}$; adaptive selection and stopping can only forgo admissions. Hence the event that some cell with H_0^c true is admitted is contained in $\bigcup_{(c,k):H_0^c} A_{c,k}$, whose probability is at most $\sum \alpha_{c,k} \le \alpha$ by Lemma 2.1 and a union bound, which needs no independence across cells. Finally, by Lemma 2.3, $\mathrm{STR}_Q(c) > -\theta + 2\xi_s$ implies $\mathrm{STR}_P(c) > -\theta$, that is, H_0^c. □

Lemma 2.3 (Shift). *If f takes values in an interval of length b, then $|\mathbb{E}_Q f - \mathbb{E}_P f| \le b \cdot \mathrm{TV}(P, Q)$. Since $Z \in [-1, 1]$, $|\mathrm{STR}_Q(c) - \mathrm{STR}_P(c)| \le 2\,\mathrm{TV}(P_s, Q_s) \le 2\xi_s$, and the constant 2 is attained.*

Proof. Centre f so that $|g| \le b/2$; then $|\mathbb{E}_Q g - \mathbb{E}_P g| \le (b/2) \sum_x |Q(x) - P(x)| = b\,\mathrm{TV}(P, Q)$. For attainment, take task types u with $Z = 1$ and d with $Z = -1$, $P(u) = 0.1$, $Q(u) = 0.2$: then TV $= 0.1$ and the risks differ by 0.2. □

Why the budget is per cell. Allocating one α_s per *scope* and letting every skill in the scope use the threshold $1/\alpha_s$ does not control the family-wise error: with ten skills at the null boundary, $\theta = \alpha = 0.1$ and 300 trials, the family-wise false admission rate of our e-process is 0.45, against 0.05 with the per-cell budget $\alpha/10$ (Section 2.6).

2.3 Scope and shift are necessary

Assumption 3 (Rich target class). *For every value of the features a procedure observes (fingerprints, predicates, trial outcomes on the observed repositories), there is a repository with those features whose task losses are arbitrary in* $[0, 1]$.

Proposition 2.4 (No guarantee without target evidence or bounded shift). (a) *Let a procedure decide from data of observed repositories* $\mathcal{R}_{\text{obs}}$ *only, and admit a skill* S *with positive probability. Under Assumption 3 there is a repository* $R^\dagger \notin \mathcal{R}_{\text{obs}}$ *with* $\mathrm{STR}(S, \{R^\dagger\}) = 1$ *on which the admission probability is unchanged.* (b) *For every* $\theta \in (0, 1)$, $\delta > 0$ *with* $\theta + \delta < 1$, $\xi \in (0, 1/2]$ *and every procedure, there are* P, Q *with* $\mathrm{TV}(P, Q) = \xi$ *and a cell* c *with* $\mathrm{STR}_P(c) = -\theta - \delta$ *and* $\mathrm{STR}_Q(c) = \mathrm{STR}_P(c) + 2\xi$*, whose admission probability does not depend on* Q.

Proof. (a) Take $R^\dagger$ that agrees with all observed data and has $\ell(x, \pi) = 1$, $\ell(x, \varnothing) = 0$ for every task. (b) Take task types u ($Z = 1$) and d ($Z = -1$) with $P(d) = (1 + \theta + \delta)/2 \geq \xi$, and obtain Q by moving mass ξ from d to u. The procedure only reads tasks drawn from P. □

Part (b) shows that no procedure that learns only from P can certify Q-risk better than $\mathrm{STR}_P + 2\xi$, so the shift term of Theorem 2.2 cannot be removed or reduced; this is why the guarantee is relative to a declared scope family and why coarse scopes make it weak.

2.4 Composition

Theorem 2.5 (Composition risk is not controlled by individual risks). *For* $a, b \in [-1, 0)$,

$$\sup\{\mathrm{STR}(S_1 \circ S_2) : \mathrm{STR}(S_1) = a,\ \mathrm{STR}(S_2) = b\} = 1 + \min(a, b),$$

and the supremum is attained. In particular, two skills that each meet the margin $-\theta$ *can compose to excess risk* $1 - \theta > 0$*, and no bound on the composite risk improves on the trivial one implied by* $\ell \in [0, 1]$.

Proof. Upper bound: $\mathrm{STR}(S_1 \circ S_2) \leq 1 - \mathbb{E}\ell(\varnothing)$, and $\mathbb{E}\ell(\varnothing) \geq -a$ because $\mathbb{E}\ell(\pi_1) = \mathbb{E}\ell(\varnothing) + a \geq 0$; likewise $\geq -b$. Attainment: constant losses $\ell(\varnothing) = -\min(a, b)$, $\ell(\pi_1) = \ell(\varnothing) + a$, $\ell(\pi_2) = \ell(\varnothing) + b$, $\ell(\pi_1 \circ \pi_2) = 1$. □

Theorem 2.5 says nothing about footprints. Screening only pairs whose footprints overlap is therefore sound only under an additional assumption, which we state and test rather than presume.

Assumption 4 (Non-interaction of disjoint footprints). *If* $\mathcal{K} \cap \mathcal{K}' = \varnothing$ *then* $\ell(x, \pi \circ \pi') - \ell(x, \varnothing) = [\ell(x, \pi) - \ell(x, \varnothing)] + [\ell(x, \pi') - \ell(x, \varnothing)]$ *for all* x.

Corollary 2.6. *Under Assumption 4, two admissible skills with disjoint footprints compose to* $\mathrm{STR} \leq -2\theta$*, so only footprint-overlapping pairs need a composite cell; with mean overlap degree* $\bar{d}$ *there are* $|\mathcal{L}_s|\,\bar{d}/2$ *such pairs per scope.*

2.5 The value of pairing

Proposition 2.7 (Pairing). *Let* $V_\pi = \mathrm{Var}\,\ell(x, \pi)$, $V_\varnothing = \mathrm{Var}\,\ell(x, \varnothing)$ *and* $C = \mathrm{Cov}(\ell(x, \pi), \ell(x, \varnothing))$ *on the same task. Estimating* STR *from* n *paired tasks has variance* $(V_\pi + V_\varnothing - 2C)/n$*; from two independent arms of* n *tasks each it has variance* $(V_\pi + V_\varnothing)/n$*. Both designs cost* $2n$ *runs, so pairing reduces the number of tasks needed for a given precision by* $R = (V_\pi + V_\varnothing)/(V_\pi + V_\varnothing - 2C)$*. Under the additive model* $\ell(\varnothing) = T$, $\ell(\pi) = T + S$ *with independent* T, S, $R = 1 + 2\sigma_T^2/\sigma_S^2$.

Proof. Direct computation of the two variances; under the additive model $V_\pi + V_\varnothing = 2\sigma_T^2 + \sigma_S^2$ and $C = \sigma_T^2$. □

2.6 Numerical checks

Every statement above is checked numerically by a script in the artifact, using the e-process of our implementation: per-cell validity, the failure of a per-scope budget, the attained constant 2 in Lemma 2.3, the bound of Theorem 2.5 on random instances, the pairing ratio, and the convergence of the ledger.

3 Related work (excerpt)

3.1 Risk control, e-processes and multiple testing

Conformal prediction [7], conformal risk control [2] and Learn-Then-Test [1] give distribution-free guarantees under exchangeability. Skill transfer violates exchangeability across repositories, so our guarantee is relative to a declared scope family (Proposition 2.4). For sequential evidence we use betting e-processes [9] and Ville's inequality [6], which make an admission valid at any data-dependent stopping time [4, 5]. Simultaneous control over the library is family-wise (FWER) and comes from a union bound over cells; it requires no independence. A false-discovery-rate variant is available through e-BH [8], at the price of a weaker, FDR-type guarantee; Benjamini–Hochberg [3] is used only for the secondary comparisons of the evaluation.

References

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